YANCHEN LIU

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EDUCATION HISTORY

M.Sc. in Computer Science **Sep. 2025 – Jun. 2027**
University of Southern California, LA, CA, US
B.Eng. in Computer Science **Sep. 2021 – Jun. 2025**
Shanghai University, Shanghai, CN

INDUSTRY EXPERIENCE

Software Development Engineer Intern, AI/ML **Jun. 2026 – Aug. 2026**
Amazon Lab 126 - ***Kindle*** *(San Jose, CA, US)*
Topic: Benchmark System for Agentic Coding

1. Designed and built a benchmark system for Kindle coding agents. It auto-crawls the history of internal packages into a reproducible task dataset, benchmarks each agent configuration inside an execution harness, and reports performance metrics and analysis.
2. Designed the workflow as a three-stage pipeline: 1) sample generation (provide each sample with a leak-free spec plus a held-out hidden test), 2) agent run (develop on a clean-base tree behind a read/edit/build/self-test loop), and 3) evaluation (an oracle on f2p/p2p, an advisory LLM judge, and a behavioral-equivalence rescue).
3. Accelerated the system with a workspace pool and scheduler for isolation and parallel builds; Developed an anchor algorithm to reuse caches across different dependency snapshots, which turns hundreds of cold multi-minute builds into second-level warm rebuilds.
4. Built a knowledge-update evaluation gate for a GraphRAG knowledge base. Candidate memory and experience are evaluated by multi judges and coding harness, with human-in-the-loop for the final decision.

Machine Learning Engineer Intern **Jul. 2024 - Jun. 2025**
Shanghai AI Laboratory *(Shanghai, CN)*
Topic: LLM Inference Engine, AI Compiler for Graph, Post-training

1. Extended *vllm* for *LLM inference* on in-house TPUs, including researching and adapting *Speculative Decoding*, *Paged-Attention* and *Continuous-Batching*.
2. Developed *high-performance kernels* with *MLIR* to ensure seamless compatibility and optimal performance of LLMs on in-house TPUs.
3. Conducted research on *LLM knowledge injection and fine-tuning* for *kernel fusion and translation* across different hardware platforms.

RESEARCH EXPERIENCE

Graduate Research Assistant **Jun. 2025 - Present**
University of Southern California - ***INK Lab*** *(Los Angeles, CA, US)*
Topic: RLHF and Reasoning for LLMs

1. Led the research of LLM-based game-playing agents, using replay-anchored judge as the long-horizon reward signal in reinforcement learning; Developed a text-based game-playing harness; Patched slime for training.
2. Led the research of user simulation analysis, designed a benchmark system for individual fidelity of user simulators; Proved that current methods are mixing individualization with population and recency.
3. Participated in the research of LLM-based recommendation agent; Proposed a new token-level self-distillation method for agentic RL on different recommendation tasks.
4. Proposed Soft-Thinking Policy Optimization, which explores the paradigm of LLM's latent-reasoning, including patching *vllm* and *sglang*, building an optimized Tree-Seach generation and Training engine.
5. Enhanced *Chain-of-Thought* data with *segment and token importance* evaluated by different methods; Utilized the data for Supervised Fine-Tuning of various reasoning models with *unsloth*.

Undergraduate Research Assistant **Mar. 2023 - Apr. 2025**
Shanghai University Cyber Security Lab *(Shanghai, CN)*
Topic: Vehicle Modeling, Simulation, and Intrusion Detection

1. Deep learning research on cyber security and functional safety in *in-vehicle communication system*.
2. Developed and optimized a novel *gradient descent solver* for *Multi-Dimensional Hawkes Process*; Introduced *MDHP-Net* for improved timing feature extraction in in-vehicle communication data.

Undergraduate Research Intern **Jun.2023 – Jul. 2024**

Shanghai University & ***East China Air Traffic Control Bureau*** *(Shanghai, CN)*

Topic: Super Resolution for Meteorological Data

1. Deployed DL models including *PanGu* and *FourCastNet* to improve *meteorological monitoring, forecasting, and super-resolution* for the East China Air Traffic Control Bureau.
2. Utilized *key-point and semantic constraints*, combined with *feature map fusion*, to improve the up-sampling process in *super-resolution* models.

HONORS AND AWARDS

1. [2024] *First Prize* and *Group Competition Award* in 2024 ASC Student Supercomputer Challenge Global Final.
2. [2022] *First-Class Academic Scholarship* for outstanding academic performance, Shanghai University.

PUBLICATIONS

2026 ------

1. [ICLR Poster] S. Wang, **Y. Liu**, X. Ren. "*Segment-Level Attribution for Selective Learning of Long Reasoning Traces*".
2. **Y. Liu†**, X. Che†, B. Joshi†, R. Fukuda, J. Liu, R. Li, Y. Ge, H. Yan, X. Ren. "*Replay-Grounded Signals for Long-Horizon Civilization Agents*".
3. **Y. Liu†**, J. Liu†, B. Joshi†, X. Che, R. Li, X. Ren. "*Attribution Trials: Auditing Whether User Simulators Represent the Intended Individual*".
4. B. Joshi†, **Y. Liu†**, S. Swayamdipta, X. Ren, T. Paek, R. Koncel-Kedziorski, "*Learning to Seek User Information for Personalization via Multi-turn Conversations*".
5. J. Liu†, **Y. Liu†**, L. He, Z. Zhang, X. Che, L. Ge, F. Qian, "*When Repairs Stop Being Recipes: Re-Grounding Repair Memory for Coding Agents*".
6. X. Che, H. Yan, **Y. Liu**, Y. Ge, H. Liu, A. Shi, R. Li, H. Wang. "*Does Learning to Predict the World Help LLM Agents Act? Auditing World-Model Post-Training*".
7. X. Che, S. Li, Y. Ge, **Y. Liu**, H. Yan, X. Lei, Y. Wang, Z. Dong, Y. Yao, Q. Xie, J. Liu. "*GameLogicBench: Evaluating Coding Agents on Runtime Game Logic with Frame-Level Behavioral Assertions*".

2025 ------

8. J. Lv, X. He, **Y. Liu**, A. Shen, X. Dai, Y. Li, J. Hao, J. Ding, Y. Hu, S. Yin. "*HPCTransEval: A Benchmark of High-Performance GPU-to-CPU Transpilation with Pre-trained Large Language Models*".
9. Q. Liu†, **Y. Liu†**, R. Li, C. Cao, Y. Li*, X. Li*, P. Wang, R. Feng, "*MDHP-Net: Detecting an Emerging Time-Exciting Threat in IVN*".

2024 ------

10. [IEEE Internet of Things Journal] Q. Liu, X. Li, K. Sun, Y. Li* and **Y. Liu***, "*SISSA: Real-Time Monitoring of Hardware Functional Safety and Cybersecurity With In-Vehicle SOME/IP Ethernet Traffic*".
11. [MDPI Future Internet] Li, X., R. Li, and **Y. Liu**. "*HP-LSTM: Hawkes Process-LSTM-Based Detection of DDoS Attack for In-Vehicle Network*".

TECH STACK

Programing Languages

1. C++ (cmake, bazel, vcpkg, gcc, clang, msvc, msbuild, simd)
2. CUDA (simt, nsight compute)
3. Python (pip, conda, uv)

AI/ML

1. **Post-Training & RL** (lora, sft, opd, rlhf, ppo, grpo, dapo, opsd, sdpo, agentic rl; trl, verl, slime)
2. **Inference Acceleration** (TP, prefilling, decoding, prefix caching, continuous-batching, Chunked Prefill, Speculative Decoding, Quantization, Paged Attention, Flash Attention, MQA, GQA, MLA)
3. **Training Acceleration** (Mixed Precision, Gradient Checkpointing, ZeRO, DP, DDP, FSDP, MP, PP)

Dev Tools

1. Git, GitHub, GitLab, Brazil (Amazon)
2. Docker, Apptainer, Singularity, Slurm, Module
3. Linux (Ubuntu, Debian, Fedora, Rocky Linux), WSL2